Light

GCSE Computer Science – Computational Thinking (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is an algorithm?

- ○ **A.** A type of computer hardware component
- ○ **B.** A device that stores data permanently
- ○ **C.** A programming language used to write software
- ○ **D.** A sequence of steps that can be followed to complete a task

Q2 (1 mark)

What does the term 'decomposition' mean in computational thinking?

- ○ **A.** Removing a program from a computer

- ○ **B.** Breaking a problem down into smaller, more manageable sub-problems
- ○ **C.** Converting code from one language to another
- ○ **D.** Deleting unnecessary files from storage

Q3 (1 mark)

What is 'abstraction' in computational thinking?

- ○ **A.** Adding more detail to make a problem clearer
- ○ **B.** Removing unnecessary detail from a problem to focus on what is important
- ○ **C.** Running a program multiple times to test it
- ○ **D.** Converting a flowchart into program code

Q4 (1 mark)

Which statement correctly describes the relationship between an algorithm and a computer program?

- ○ **A.** An algorithm and a program are exactly the same thing
- ○ **B.** A program must be written before an algorithm can be designed
- ○ **C.** A computer program is an implementation of an algorithm in a specific programming language
- ○ **D.** Algorithms can only exist inside computer programs

Q5 (1 mark)

In the algorithm step 'total ← price × quantity', what type of operation is being performed?

- ○ **A.** Input - data is being received from the user
- ○ **B.** Storage - data is being saved to a file
- ○ **C.** Output - data is being displayed to the user
- ○ **D.** Processing - data is being manipulated to produce a result

Q6 (1 mark)

How does a linear search find an item in a list?

- ○ **A.** By checking each item in the list one by one from the start until the item is found or the end is reached
- ○ **B.** By jumping to the middle of the list and eliminating half each time
- ○ **C.** By sorting the list first and then finding the item
- ○ **D.** By checking only the first and last items in the list

Q7 (1 mark)

What must be true about a list before binary search can be used?

- ○ **A.** The list must contain only numbers
- ○ **B.** The list must have fewer than 100 items

- ○ **C.** The list must contain no duplicate values
- ○ **D.** The list must be sorted in order

Q8 (1 mark)

What does bubble sort do during each pass through a list?

- ○ **A.** It divides the list into two halves
- ○ **B.** It finds the smallest item and moves it to the start
- ○ **C.** It compares adjacent pairs and swaps them if they are in the wrong order
- ○ **D.** It removes duplicate values from the list

Q9 (1 mark)

What approach does merge sort use to sort a list?

- ○ **A.** It compares adjacent pairs and swaps them repeatedly
- ○ **B.** It finds the largest item and moves it to the end
- ○ **C.** It divides the list into smaller sublists, sorts them, and merges them back together
- ○ **D.** It randomly shuffles the list until it is sorted

Q10 (1 mark)

In a flowchart, what does a diamond shape represent?

- ○ **A.** The start or end of the algorithm
- ○ **B.** A decision point where a condition is tested and the flow branches based on the result
- ○ **C.** A process or calculation step
- ○ **D.** An input or output operation

Q11 (1 mark)

What is the purpose of a trace table?

- ○ **A.** To track how variable values change as an algorithm executes step by step
- ○ **B.** To design the user interface of software
- ○ **C.** To calculate how much memory a program needs
- ○ **D.** To write the final version of a program

Q12 (1 mark)

When comparing the efficiency of two algorithms that solve the same problem, what does 'time efficiency' refer to?

- ○ **A.** How quickly the algorithm completes its task relative to another
- ○ **B.** The physical time of day the algorithm runs best
- ○ **C.** How much electricity the algorithm uses

- ○ **D.** How long it takes to type the algorithm's code

Q13 (2 marks)

A list contains 50 items. In the worst case, how many comparisons would a linear search need to make to determine that a specific item is not in the list?

- ○ **A.** 25 comparisons - the algorithm checks half the items on average
- ○ **B.** 50 comparisons - every item must be checked before concluding the item is not present
- ○ **C.** 6 comparisons - the algorithm divides the list repeatedly
- ○ **D.** 1 comparison - the algorithm can immediately determine absence

Q14 (2 marks)

A sorted list contains 1000 items. Approximately how many comparisons would binary search need in the worst case to find an item or determine it is not present?

- ○ **A.** 10 comparisons
- ○ **B.** 500 comparisons
- ○ **C.** 100 comparisons
- ○ **D.** 1000 comparisons

Q15 (2 marks)

A list of 8 items needs to be sorted using bubble sort. If the list is already sorted, what is the minimum number of passes through the list needed to confirm this?

- ○ **A.** 1 pass - if no swaps occur during a pass, the list is confirmed sorted
- ○ **B.** 8 passes are always required regardless of initial order
- ○ **C.** 4 passes - half the list length
- ○ **D.** 0 passes - bubble sort can detect sorted lists without any passes

Q16 (2 marks)

During the first phase of merge sort, a list of [8, 3, 5, 1] is repeatedly divided. What sublists exist after all divisions are complete?

- ○ **A.** [[8, 3], [5, 1]]
- ○ **B.** [[1, 3, 5, 8]] - the list is sorted immediately
- ○ **C.** [[8], [3], [5], [1]] - the list is divided until each sublist contains exactly one element
- ○ **D.** [[8, 3, 5, 1]] - no division occurs

Q17 (2 marks)

A games company stores player high scores in a list that is constantly updated with new scores being added. Why might linear search be more practical than binary search for finding a specific score?

- ○ **A.** Because linear search works on unsorted lists, avoiding the need to re-sort after every new score is added

- B. Because linear search is always faster than binary search
- C. Because binary search cannot find numbers, only text
- D. Because linear search uses less electricity

Q18 (2 marks)

Why is merge sort generally more efficient than bubble sort for sorting large lists?

- A. Merge sort uses simpler code that runs faster
- B. Bubble sort cannot handle lists larger than 100 items
- C. Merge sort divides the problem repeatedly, dramatically reducing the number of operations needed compared to bubble sort's repeated passes
- D. Merge sort requires no comparisons between elements

Q19 (2 marks)

A student needs to create a program that manages a school library: tracking books, handling loans, managing member accounts, and generating overdue notices. How would decomposition help with this task?

- A. By copying code from existing library programs
- B. By removing the overdue notice feature to simplify the program
- C. By writing all the code in a single file

○ **D.** By breaking the problem into sub-problems like 'book database', 'loan system', 'member management', and 'notification system', each of which can be developed and tested separately

Q20 (2 marks)

A programmer is creating a simulation of traffic flow in a city. Which of the following best demonstrates abstraction?

○ **A.** Including every building, tree, and street lamp in the simulation

○ **B.** Using only a high-level language to write the program

○ **C.** Running the simulation at night when there is less traffic

○ **D.** Representing cars as simple moving objects with position and speed, ignoring details like colour, make, model, and driver appearance

Q21 (2 marks)

Algorithm A takes 5 seconds to sort a list of 1000 items. Algorithm B takes 2 seconds to sort the same list. Algorithm A's code is shorter and easier to understand. Which statement about efficiency is correct?

○ **A.** Both algorithms are equally efficient because they produce the same result

○ **B.** Algorithm A is more time-efficient because shorter code always runs faster

○ **C.** Algorithm B is more time-efficient because it completes the sorting task faster, regardless of code complexity

○ **D.** Efficiency cannot be compared because the algorithms are different

Q22 (2 marks)

Consider the algorithm:

```
x ← 3
y ← 5
x ← x + y
y ← x - 2
What are the final values of x and y?
```

- ○ **A.** x = 3, y = 5
- ○ **B.** x = 8, y = 3
- ○ **C.** x = 5, y = 8
- ○ **D.** x = 8, y = 6

Q23 (2 marks)

A flowchart contains a diamond with 'Is score ≥ 50?' inside it. The 'Yes' path leads to a box containing 'OUTPUT "Pass"'. The 'No' path leads to a box containing 'OUTPUT "Fail"'. What happens when score is 50?

- ○ **A.** Nothing is output because 50 is on the boundary
- ○ **B.** The program outputs 'Fail' because 50 is not greater than 50
- ○ **C.** The program outputs 'Pass' because 50 is greater than or equal to 50
- ○ **D.** The program outputs both 'Pass' and 'Fail'

Q24 (3 marks)

A sorted list contains: [5, 12, 18, 24, 31, 38, 45, 52]. Binary search is used to find the value 18. Which sequence correctly describes the comparisons made?

- ○ **A.** Check 5, then 12, then 18 - found at position 2
- ○ **B.** Check 24 (middle), then 12 (middle of left half), then 18 (middle of right portion) - found at position 2
- ○ **C.** Check 52, then 38, then 18 - found at position 2
- ○ **D.** Check 18 immediately - found at position 2

Q25 (3 marks)

The list [6, 2, 8, 4] is sorted using bubble sort. After the first complete pass (comparing and swapping adjacent pairs from start to end), what does the list look like?

- ○ **A.** [2, 4, 6, 8] - the list is completely sorted after one pass
- ○ **B.** [2, 6, 4, 8] - the largest value has reached the end
- ○ **C.** [6, 2, 4, 8] - only the last two elements were swapped
- ○ **D.** [2, 6, 8, 4] - elements were sorted but in wrong order

Q26 (3 marks)

A teacher needs to sort a class list of 25 students by surname. The list is already almost sorted - only two students are in the wrong positions. Which sorting algorithm would likely be most efficient and why?

- ○ **A.** Merge sort - because it is always the fastest sorting algorithm
- ○ **B.** Linear search - because it can find the misplaced students quickly

○ **C.** Bubble sort - because it can detect the nearly-sorted state quickly, potentially completing in very few passes since most elements are already in position

○ **D.** A random sorting algorithm - because small lists do not matter

Q27 (3 marks)

A development team is creating a mobile app for ordering takeaway food. Using decomposition, identify which breakdown of the problem into sub-problems is most appropriate.

○ **A.** The team should focus only on making the app look attractive, ignoring other features

○ **B.** Sub-problems: choosing app colours; selecting fonts; deciding icon shapes - focusing on visual elements only

○ **C.** The app should be written entirely by one programmer to maintain consistency

○ **D.** Sub-problems: user account system (registration, login, profile); restaurant browsing (search, filters, menus); ordering process (basket, checkout, payment); delivery tracking (real-time updates, estimated times); review system (ratings, written reviews) - each can be developed separately then integrated

Q28 (3 marks)

An online dictionary contains 50,000 words stored in alphabetical order. Users search for word definitions. Which search algorithm should be used and why?

○ **A.** Binary search - because the data is already sorted alphabetically, allowing each comparison to eliminate half the remaining words, finding any word in approximately 16 comparisons instead of up to 50,000

○ **B.** Neither algorithm - dictionaries cannot be searched by computers

○ **C.** Linear search - because it is simpler to implement and understand

○ **D.** Bubble sort - because the words need to be sorted before searching

Q29 (3 marks)

A streaming music service needs to sort 10 million songs by play count daily to update its popularity charts. Which sorting approach is most appropriate and why?

○ **A.** No sorting needed - popularity charts do not require sorted data

○ **B.** Bubble sort - because it uses less memory than merge sort

○ **C.** Manual sorting - because computers cannot handle 10 million items

○ **D.** Merge sort - because its divide-and-conquer approach maintains consistent efficiency even with 10 million items, and its predictable performance is essential for a time-critical daily task that must complete reliably

Q30 (3 marks)

A programmer is asked to create software that finds the cheapest flight between two cities, considering multiple airlines, layovers, and varying prices. The programmer has never built anything similar before. What should be their first steps using computational thinking?

○ **A.** Start writing code immediately to avoid wasting time on planning

○ **B.** Use decomposition to break down the problem (data collection, route finding, price comparison, result display), use abstraction to focus on essential factors (price, duration, connections) while ignoring irrelevant details (aircraft colour, pilot names), then design algorithms for each sub-problem before writing any code

○ **C.** Give up because the problem is too complex

○ **D.** Copy an existing flight booking website entirely

Total: **55** marks • Time: **30 min**

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